Model Risk Factory

EXL × UCD · DATA & AI HACKATHON 2026

Document, migrate and validate legacy credit models



TEAM DETAILS

Team name	[Enter Team Name]
Team leader name	fangxunwu (25 208 508)
Team members (2–4)	fangxunwu, kaiwenyao
UCD email (team leader)	[Enter Team Leader Email]
Challenge selected (01–04)	04 · Model risk factory

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Problem Statement & Opportunity

THE PROBLEM

• ~250 bank models need current documentation and validation (PRA SS1/23).

• One validation takes 8–12 weeks; documents drift from the code.

• Users: model owners, validators and sign-off reviewers.

(Client context from the challenge brief.)

OPPORTUNITY & USP

01 Every fact traced to file, line and SHA-256

02 Agent drafts; a human approves each gate

03 Migration proven by parity, not retraining

Solution Overview & Key Features

BRIEF ABOUT THE IDEA

An Agent documents legacy R models, ports one to Python with a parity proof and writes the validation report. A human approves every gate.

FEATURES OFFERED BY THE SOLUTION

01
Inventory
Gaps kept, hashed

02
R → Python
Row-by-row parity

03
Validation
Perf · drift · bias

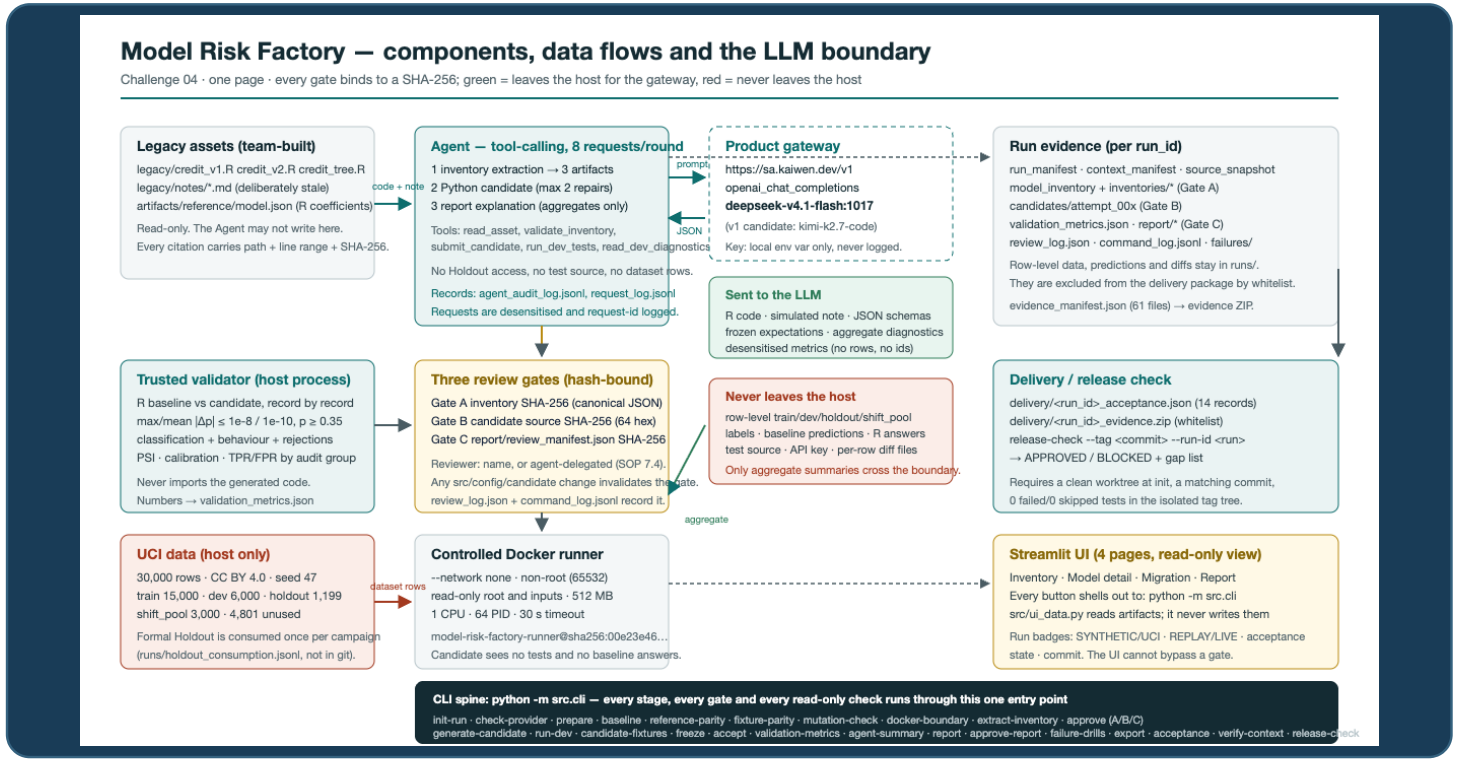
04
Human gates
Hash-bound approvals

PROCESS FLOW / USE-CASE DIAGRAM

Document → Gate A → Port to Python → Gate B → Docker parity → Holdout → Validate → Gate C → Monitor drift

Architecture, Technology & Responsible AI

ARCHITECTURE DIAGRAM



TECHNOLOGIES USED

Python · R · Docker · Streamlit

LLM: deepseek-v4.1-flash

ESTIMATED COST

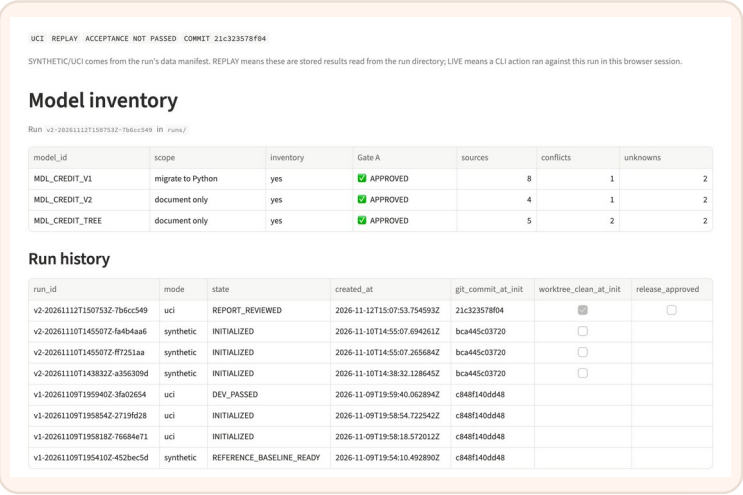
Local laptop; LLM tokens only

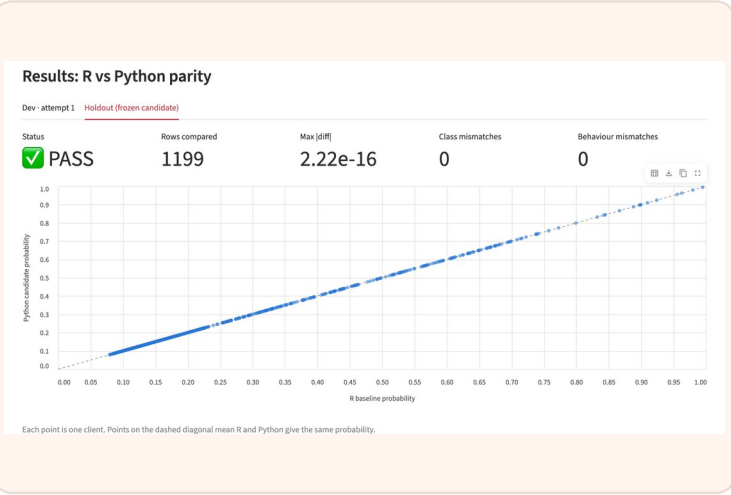
RESPONSIBLE AI · REQUIRED

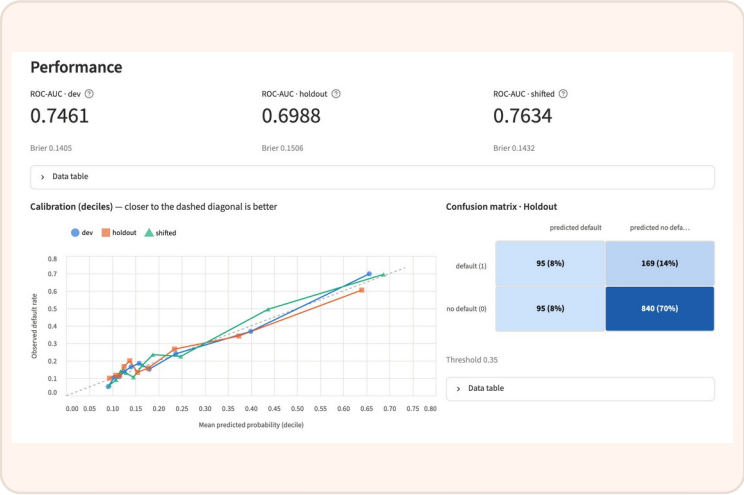
- Public data only
- No raw rows sent to LLM
- Prototype; not certified

Prototype & Performance

SNAPSHOTS OF THE PROTOTYPE







PERFORMANCE / BENCHMARKING

2.2e-16 — R vs Python gap, 1,199 rows

3.5 min — Agent pipeline (synthetic)

0.699 — Holdout AUC, seed 47

Links, Future Development & How to Submit

PROVIDE LINKS TO YOUR

GitHub public repository
github.com/wufangxun106334-pixel/model-risk-factory

Demo video link (≤ 3 min)
Link pending (1:51 MP4 ready)

Final product / deployed link
Local Streamlit app; not deployed

AI tools / existing code used
Claude Code, Kimi Code, deepseek-v4.1

FUTURE DEVELOPMENT

13/14 checks pass; release BLOCKED (D-01)

Next: SAS inputs, ~250-model queue

HOW TO SUBMIT

SUBMIT TO	Hackathon submission form →
SUBJECT	EXL Hackathon | [Team name] | Challenge 04
DEADLINE	13 Nov 2026, 23:59 Irish time*
SEND FROM	Leader's UCD email

Questions? monika.rawal@exlservice.com